

Do in-silico metrics filter designed enzymes against catalysis?

Supplementary Information

1 About these tables

This document describes the 49 supplementary tables (S1 to S49) and the seven data tables of the Results (T5 to T11) of the paper. It does not reprint them. Every one of these tables is released as a CSV file (Tables 1 to 4, in the Methods, are descriptive and are not released as files); each entry below gives the path, the number of rows and columns, what a row is, what the key columns mean, a three-row excerpt, the places in the paper that cite the table, and caveats. Supplementary tables are numbered in the order in which the Methods and Results first cite them.

README.md	what is released, what was changed, licence
INDEX.csv	one row per table: id, file, title, rows, columns, licence flag, sha256
main/	T5 ... T11 Tables 5 to 11 of the paper
supplementary/	S1 ... S49 supplementary tables, one CSV per table

How to read an entry. *File* is the path of the CSV, its size and its licence flag. *Supports* is the part of the paper the table backs. *Cited in* lists the sections of the Methods, Results and Discussion that cite it (a link opens the paper at that section). *A row is* says what one row represents. *Key columns* explains the columns that matter; the shared provenance block (below) is not repeated. The *excerpt* shows real rows from the file.

2 Conventions shared by the tables

Shared provenance columns. The result tables (T5 to T11 and the supplementary tables whose entry says so) carry the same block of up to 12 columns, so that a row can be traced to its denominator: `claim_type`, `unit` and `n_units` (what was counted and how many), `n_clusters` and `cluster_def` (the resampling unit: enzyme sub-subclass for natural enzymes, campaign and plate well or backbone cluster for designs, position for BglB), `k`, `N` and `k_of_N` (a count with its denominator), `denominator_def`, `headline_ok` (1 if the row may be quoted as a headline result; 0 for per-metric detail, descriptive analyses, reference rows and wrong-ligand AUROCs), `status` and `licence_flag`.

Claim types. `detection` (can the metric see damage), `specificity` (catalytic against matched control), `equivalence` (panel ratio against 1), `discrimination` (a real contrast, dead against active), `ranking` (against a measured label), `enrichment`, `baseline` (a reference row), `design`, `audit`, `not_measurable`.

Applicability status (in S3, S4 and the `applicability_status` columns). OK can be tested; OK-KNOCKON tested but reported as displacement of the other catalytic residues, with no specificity verdict; OK-PROVISIONAL the raw status of the interface terms in the re-predicted ladder, resolved by testability; GLOBAL tested but has no site input, so it is a comparator; REF a fixed reference row; X-BOOKKEEP a counter or constant; X-NOINPUT the input cannot reach the changed quantity; X-ARM the metric belongs to the other kind (structure space or prediction) than the test; X-UNDEF-COV defined on too few units; X-WITHDRAWN the test was withdrawn; X-NOCTRL no matched control exists; X-CONFOUND the control is confounded; NOT-RUN; DUP identical to another test and read once.

Verdicts in the specificity tables. `specific` (response significant, interval of the specificity ratio above 1, control adequate); `non_specific` (responds, not more than the control); `blind` (does not respond); `invariant` (does not change); and counts-floor labels for cells with too few units.

Why some metrics are not in the main set. A metric with no residue-set input (whole-protein energies, sequence composition, whole-prediction confidence) cannot respond *specifically* to catalytic residues, and a metric whose input cannot reach the changed quantity cannot register the lesion at all (sequence composition under a rotamer change, for example); such metrics are

kept as comparators. One further quantity passes these tests but is kept out of the main set: the Chai-1 combined score, which Chai-1 uses to rank its own models and which equals $0.2 \times \text{pTM} + 0.8 \times \text{ipTM} - 100 \times (\text{inter-chain clash flag})$, so that it is a rescaled ipTM and not an independent metric. It is still scored, and its role and reason are in [S1](#).

Names that recur. `canonical_key` is the distinct metric after literal aliases are merged (`metric_name` may be an alias); `in_main_set` is 1 for the 33 reported metrics; `role` is main, reference or a supplement role (see [S1](#)); `source` is structure-space, prediction-based, PLACER, baseline, natural or denovo as the table says; `lesion_test` is the catalytic lesion or the second-shell lesion; `level` or `lesion_step` is the lesion step (isosteric 15.3, non-isosteric 36.4, Ala 53.2, Gly 80.5 Å³ median side-chain volume change; `second_shell_1/2/4` is the second-shell dose); `ame_flavour` is crystal (against the deposited structure) or `self_design` (against a design's own model) and the two are never pooled; `sr_*` is a specificity ratio with its interval; `auroc` is direction-free unless a column says signed or directed; `rho` is a Spearman correlation.

Test names. Tables that refer to a test use the names of [S6](#), a name followed by its class in square brackets, for example main set [catalytic lesion, structure-space].

Licence. Tables flagged `D2DCure_aggregate` in `INDEX.csv` are built from the D2DCure BglB data, whose licence is unstated; they hold metric-level aggregates only, no per-variant row. All other tables are public.

Reproducibility. The tables were generated by script from result files with seeds and input hashes recorded; `INDEX.csv` gives the sha256 of every released file.

3 Where do I find ...?

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Which 33 metrics are reported, and why the rest are not	S1 , S2 ; then S3 and S4 for the reason in each test
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The 30 tests, with comparator, control type and role	S6
The specificity ratio of one metric at one lesion step	S33 (comparators: S34)
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How fragile the structure-space specificity result is	S31 , S42 , S29
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Every metric on the 192 plated designs, active against no active (complete Table 6)	S13
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Every metric on the 16 plated designs with a measured activity (complete Table 9)	S21
Hits per plate when a metric fills a 96-well plate	S20
The wider set of 49 pairs, the 28 re-predicted pairs, the pair-set definitions	S7 , S8 , S9
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S13	Plated designs: all 128 metrics ranked by the AUROC of active against no active designs (complete Table 6)	Activity discrimination (3.1)	128 × 29	11
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S18	BglB: sensitivity analyses	Activity ranking (3.3)	11 × 22	14
S19	BglB: prediction-based metrics (Chai-1 with the assay substrate)	Activity ranking (3.3)	28 × 29	15
S20	Plated designs: hits per 96-well plate for every metric (filter-style analysis)	Activity ranking (3.3)	128 × 37	16
S21	Plated designs: every metric ranked by correlation with kcat/KM among the 16 designs that have a value (complete Table 9)	Activity ranking (3.3)	106 × 29	17
S22	Plated designs: ordering among the 16 active designs, structure-space metrics and comparators	Activity ranking (3.3)	81 × 23	17
S23	Plated designs: enrichment at every keep fraction, structure-space metrics	Activity ranking (3.3)	816 × 27	18
S24	Plated designs: enrichment for prediction-based metrics	Activity ranking (3.3)	232 × 27	18
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S26	Control matching: the loosest tier needed	Detection ability (3.4)	81 × 6	19
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S29	Structure-space specificity by metric family and lesion step, with the panel statistic and sensitivity sets	Detection ability (3.4)	20 × 27	21
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S36	Rank statistic R per lesion step on the 143 main enzymes	Detection ability (3.4)	313 × 14	25
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S40	Re-predicted ladder: caliper-matched subsets and the rule labels	Detection ability (3.4)	348 × 25	27
S41	Re-predicted ladder: regression-adjusted ratios	Detection ability (3.4)	116 × 28	28
S42	Panel equivalence statistic: earlier whole panel against the main-set metrics	Detection ability (3.4)	6 × 10	28
S43	Re-predicted ladder: specificity ratio per step for all metrics	Detection ability (3.4)	231 × 16	29
S44	Second-shell lesion: every metric scored (Table 11 plus supplementary comparators and per-dose flags)	Detection ability (3.4)	37 × 59	29
S45	Drift of the current Chai-1 engine against the earlier predictions	Substrate discrimination (3.2)	176 × 5	30
S46	Effective dimensionality of the panel	Detection ability (3.4)	4 × 10	31
S47	De novo designs: burial equivalence of the created-site control	Detection ability (3.4)	5 × 11	31

Table	Title	Topic	Rows × cols	Page
S48	Experiments that could not be made or were retired	Detection ability (3.4)	9 × 4	32
S49	PLACER ensemble metrics on the isosteric step (not interpretable)	Detection ability (3.4)	50 × 12	32

5 Supplementary tables

Table S1. Main metric set: the role of every distinct metric, and why

File. supplementary/S1_main_metric_set.csv · 191 rows × 16 columns · public

Supports. Methods, Table 1 and 'Which metrics are reported'; the main set of 33 metrics used in Tables 5 to 11. **Cited in.** Methods 2.1.

A row is one distinct metric (a canonical key; literal aliases are merged), 191 in all.

The audit that fixes which metrics the paper reports. For each distinct metric it gives the applicability status in the detection test of the catalytic lesion, in the dead-against-active tests and in the activity-ranking tests, and the resulting role: main (33, valid and tested in the deliberate-change tests and in the tests on real proteins), reference (6, comparators chosen before any result was seen), excluded_bookkeeping (63 counters and constants) or one of the supplement roles (supp_global 44, supp_placer 25, supp_excluded 10, supp_aggregate 4, supp_partial 3, supp_real_protein_tests_only 2, and supp_derived 1, the Chai-1 combined score, a rescaled ipTM). The reason for each role is in the last column. The status is decided from what a metric reads, whether it is defined and the design of the experiment, never from a result.

Key columns.

Column	Meaning
canonical_key / members / rep_metric	the distinct metric, the names merged into it and the representative one
mode / input_class / subfamily / aggregator	structure-space or prediction-based, what the metric reads, its family and how it is summarised
status_detection_test / fraction_detection_test	applicability status in the detection test of the catalytic lesion and the share of units on which the metric is defined
status_dead_vs_active_tests / fraction_dead_vs_active_tests / partial_dead_vs_active_tests	the same for the dead-against-active tests
status_ranking_tests / ranking_tests_ok	status in each activity-ranking test (BglB variants, plated designs) and the tests in which the metric is OK
role	main, reference, excluded_bookkeeping, supp_global, supp_placer, supp_excluded, supp_aggregate, supp_partial, supp_real_protein_tests_only or supp_derived
reason	why the metric has that role

Excerpt (first 3 rows where role = main).

canonical_key	input_class	status_detection_test	status_dead_vs_active_tests	role
ame_crystal_clash_max	pred_interface	OK-PROVISIONAL	OK	main
ame_crystal_rmsd	ame	OK-KNOCKON	OK	main
catalytic_frac_buried	struct_site	OK	OK	main

Notes.

- The status vocabulary is in the Conventions.
- NOT-RUN in status_ranking_tests marks the PLACER metrics, which were not computed in the BglB and plated-design analyses.

Table S2. Metric contracts: what each named quantity reads, over which residues, and where it is valid

File. supplementary/S2_metric_contracts.csv · 216 rows × 19 columns · public

Supports. Methods, Table 1 and 'Which metrics are reported'. **Cited in.** Methods 2.1.

A row is one named quantity (216 in all; an alias is a separate row linked by alias_of and canonical_key).

The input contract of every quantity in the panel: what it reads (sequence only, whole-protein structure, site-scoped structure, predictor output, active-site accuracy, PLACER output, deposit metadata or bookkeeping), over which residues it is scored in the catalytic arm and in the two kinds of control arm, which inputs it needs, which kinds of metric it is valid for and which changes it cannot respond to.

Key columns.

Column	Meaning
metric_name / canonical_key / alias_of / alias_kind	the quantity, the distinct metric after aliases are merged, and the kind of alias
aggregator / metric_family / subfamily	how it is summarised, and its family and subfamily
input_class	sequence-only, whole-protein structure, site-scoped structure, prediction, active-site accuracy (AME), PLACER, deposit metadata or bookkeeping
scope_catalytic_arm / scope_control_arm_catalytic_lesion / scope_control_arm_second_shell	the residues the metric is scored over in the catalytic arm, in the control arm of the catalytic lesion and in the control arm of the second-shell lesion
needs_cat_residue / needs_ligand / needs_reference / needs_predictor	the inputs the metric needs
valid_modes	structure-space, prediction-based, PLACER, any or deposit metadata
invariant_under	changes the metric cannot register (for example sequence composition under a rotamer change)
ame_flavour / note	for active-site accuracy, the reference it is measured against; remarks on the metric, including where the metric as computed differs from its earlier description

Excerpt (first 3 rows where input_class = struct_site).

metric_name	input_class	scope_catalytic_arm	valid_modes
catalytic_frac_buried	struct_site	scored_residue_set	structure-space
catalytic_ligand_min_dist	struct_site	scored_residue_set+ligand_within_...	structure-space
catalytic_pairwise_dist_max	struct_site	scored_residue_set	structure-space

Table S3. Applicability of every metric in every test (long format)

File. supplementary/S3_applicability.csv · 6,480 rows × 11 columns · public

Supports. Methods, 'Which metrics are reported'; every Results subsection. **Cited in.** Methods 2.1.

A row is one named quantity in one test (216 quantities x 30 tests = 6,480 rows).

Whether each quantity can be tested in each of the 30 tests, and if not why. Status counts: OK 468, GLOBAL 339, REF 15, OK-KNOCKON 6, OK-PROVISIONAL 10 (testable); X-ARM 2,296, X-BOOKKEEP 1,920, X-NOINPUT 576, X-WITHDRAWN 381, X-UNDEF-COV 203, X-NOCTRL 118, X-CONFOUND 10, NOT-RUN 79, DUP 59 (not testable, with the reason). The reason and the coverage (defined_k of defined_N units) are given. Decided from inputs, definedness and design only, never from results.

Key columns.

Column	Meaning
metric_name / canonical_key / input_class	the quantity, the distinct metric and its class
test / test_class	the test (its name and class as in S6) and the class alone, for example 'dead against active, prediction-based'
status / reason	the applicability status and the reason in words

Column	Meaning
defined_k / defined_N / fraction / partial	number of units on which the metric is defined, of how many, their ratio, and whether coverage is partial

Excerpt (first 3 rows where status = OK).

metric_name	test	status	reason
ame_crystal_clash_max	BglB single-point variants (predi...	OK	eligible and scored
ame_crystal_clash_max	zymogen-mature pairs re-predicted...	OK	eligible and scored
ame_crystal_clash_max	substrate swap, natural enzymes [...]	OK	eligible and scored

Notes.

- Released as a CSV only: with 6,480 rows it is not printed. [S25](#) is its summary by class of metric.
- Of the 79 cells with status NOT-RUN, 50 are PLACER metrics that were not computed in the BglB and plated-design analyses. The other 29 are prediction-based metrics in the ladder with ligand-distance-matched controls: this analysis was carried out (results in [S40](#) and [S41](#)) but its cells keep the label NOT-RUN.

Table S4. Applicability by class of metric and test

File. supplementary/S4_applicability_by_class.csv · 340 rows × 5 columns · public

Supports. Methods, 'Which metrics are reported' (summary of [S3](#)). **Cited in.** Methods [2.1](#).

A row is one class of metric in one test with one status (340 rows).

A summary of [S3](#): for each input class and test, how many metrics receive each status. The quickest way to see why a whole class of metrics is absent from a result.

Key columns.

Column	Meaning
input_class	class of metric (ame, bookkeeping, pred_global, pred_interface, seq_only, struct_site, struct_whole, ...)
test / test_class	the test and its class
status	applicability status
n_metrics	number of metrics of that class with that status in that test

Excerpt (first 3 rows where status = OK).

input_class	test_class	status	n_metrics
ame	activity ranking, prediction-based	OK	5
ame	dead against active, prediction-b...	OK	6
ame	substrate swap	OK	6

Table S5. Counts restated on the eligible metrics (denominators before and after the audit)

File. supplementary/S5_denominator_restatement.csv · 58 rows × 14 columns · public

Supports. Methods, 'Which metrics are reported'. **Cited in.** Methods [2.1](#).

A row is one headline count of the earlier whole-panel analysis (58 in all).

The counts of responding and specific metrics of the earlier analysis (original_k of original_N, over every metric with a testable cell) restated on the eligible metrics only (eligible_k of eligible_N, also as distinct metrics and for the whole-protein comparators), with the breakdown of what was excluded and why. It shows how much of the earlier denominators were metrics that could not respond.

Key columns.

Column	Meaning
analysis / level / statistic	the earlier analysis (data set and lesion), the lesion step and the statistic ('responds' or 'specific')
original_k / original_N	count and denominator of the earlier analysis
eligible_k / eligible_N / eligible_distinct_k / eligible_distinct_N	the same on eligible metrics, and as distinct metrics
global_k / global_N	the count among whole-protein comparators
excluded_breakdown / recompute_kind / note	which statuses were excluded and how many, how the recount was done, and a note

Excerpt (first 3 rows).

analysis	level	statistic	original_k	original_N	eligible_k	eligible_N
main set, packing-matched control...	ala	responds	55	102	25	37
main set, packing-matched control...	ala	specific	11	102	10	37
main set, packing-matched control...	gly	responds	58	98	25	34

Table S6. Test inventory: the 30 tests, with comparator, control type, role and caveats

File. supplementary/S6_test_inventory.csv · 30 rows × 14 columns · public

Supports. Methods 2.2 to 2.5; the key to the test names in [S3](#) and [S4](#). **Cited in.** Methods 2.1.

A row is one test: an experiment on a data set with one kind of metric (30 in all).

Every test with its name and class, the kind of metric (structure-space or prediction-based), the unit and the number of units, the resampling unit, the changes applied, the comparator and control type, the role (primary, sensitivity, sanity, retired, not_measurable or dup) and the caveats. [S3](#) and [S4](#) use the test names of this table.

Key columns.

Column	Meaning
test / reader_name / test_class / mode	the test (name with its class), the name alone, the class, and structure-space or prediction-based
unit / n_units / n_clusters / cluster_def	what is counted, how many, and the resampling unit
levels / comparator / control_type	what is changed or contrasted, what the metric is compared with and how the control is built
claim_type / role	the kind of claim and the role: primary, sensitivity, sanity, retired, not_measurable or dup
caveats	limits of the test

Excerpt (first 3 rows).

reader_name	test_class	mode	claim_type	role
main set, packing- matched controls	catalytic lesion, structure-space	structure-space	specificity	primary
sanity floor: all catalytic resid...	detection floor (duplicate of the...	structure-space	detection	dup
sanity floor: permuted sequence o...	sanity floor	structure-space	detection	sanity

Table S7. Wider set of 49 zymogen-mature pairs: all 136 scored metrics, with the same-state null

File. supplementary/S7_zymogen_all_metrics.csv · 136 rows × 25 columns · public

Supports. Results 3.1.1 (wider set); Methods 2.2.1. **Cited in.** Methods 2.2.1; Results 3.1.1.

A row is one scored metric (136 in all).

The analysis on the 49 pairs for which structure-space metrics were scored (before the evaluation set of 21 pairs that share a ligand was fixed): the direction-free AUROC of dead against active with interval, the directed AUROC, the number of pairs, and the AUROC of the same metric on 96 same-state null pairs, which shows how much separation arises without any dead/active difference.

Key columns.

Column	Meaning
metric_name / canonical_key / input_class / in_main_set	the metric, its class and whether it is a main-set metric
n_pairs	pairs with a value (49 at most)
auroc / auroc_lo / auroc_hi / auroc_directed	direction-free AUROC with 95% interval over pairs, and the directed value
auroc_vs_same_state_null / n_zymogen_pairs / n_same_state_null_pairs	AUROC of zymogens against the same-state null pairs and the numbers in each group
applicability_status	applicability status

The shared block of 12 provenance columns (see Conventions) is also present and is not repeated above.

Excerpt (first 3 rows).

metric_name	in_main_set	n_pairs	auroc	auroc_lo	auroc_hi
catalytic_frac_buried	1	49	0.5321	0.5019	0.6112
catalytic_hydratio_n_count	0	48	0.5686	0.505	0.6604
catalytic_ligand_m_in_dist	0	25	0.5264	0.5024	0.6464

Notes.

- Structure-space and PLACER metrics only; the 4 prediction-based metrics were scored on the 28 re-predicted pairs (S8).
- Results are descriptive of a wider set; the evaluation set is the 21 pairs of S36.

Table S8. Zymogen pairs re-predicted with Chai-1, restated with the definedness gate

File. supplementary/S8_zymogen_re_predicted_restated.csv · 33 rows × 22 columns · public

Supports. Results 3.1.1 (wider-set check for the prediction-based metrics). **Cited in.** Methods 2.2.1; Results 3.1.1.

A row is one prediction-based metric (33 in all).

The 28 zymogen-mature pairs re-predicted with Chai-1: the AUROC of each metric with interval, the number of pairs reported against the number on which the metric is defined, and a note where the earlier value was not interpretable. Active-site accuracy is relabelled as measured against each form's own deposited structure.

Key columns.

Column	Meaning
metric_name / metric_reported_as	the metric and the name it carried in the earlier report
input_class	class of metric
n_pairs_reported / n_pairs_defined_restated	pairs in the earlier report and pairs on which the metric is defined
auroc / auroc_lo / auroc_hi	direction-free AUROC with 95% interval

Column	Meaning
applicability_status / restatement_note	applicability status and what was restated

The shared block of 12 provenance columns (see Conventions) is also present and is not repeated above.

Excerpt (first 3 rows where applicability_status = OK).

metric_name	n_pairs_reported	n_pairs_defined_restated	auroc	applicability_status
ame_crystal_clash_max	21	21	0.5351	OK
ame_crystal_pass	28	28	0.5045	OK
ame_crystal_rmsd_max	28	28	0.5332	OK

Notes.

- 28 pairs include the 7 apo pairs that are not in the 21-pair evaluation set.

Table S9. Definitions of the pair sets

File. supplementary/S9_pair_set_definitions.csv · 6 rows × 4 columns · public

Supports. Methods 2.2.1; Results 3.1.1. **Cited in.** Methods 2.2.1; Results 3.1.1.

A row is one set of zymogen-mature pairs (6 in all).

The six sets used in 3.1 with their size and definition: 55 candidate rows, 49 scored pairs, 45 with PLACER, 28 re-predicted pairs, the 21-pair evaluation set (pairs that share a ligand) and the 96 same-state null pairs, and which table uses each.

Key columns.

Column	Meaning
pair_set / n	the set and its size
definition	how the set is defined
used_for	the table that uses it

Excerpt (first 3 rows).

pair_set	n	used_for
zymogen candidate rows	55	candidate list
scored pairs (wider set)	49	S7 (49 = max n_pairs)
pairs with PLACER	45	S49

Table S10. Substrate-trapping mutants checked against the literature (tier retired)

File. supplementary/S10_trapping_verification.csv · 6 rows × 4 columns · public

Supports. Results 3.1.1 (retired tier); Methods 2.2.1. **Cited in.** Methods 2.2.1; Results 3.1.1.

A row is one verdict category (6 rows).

The 30 trapping-mutant pairs verified against the primary literature: 15 confirmed reduced, 4 confirmed inactive, 4 active and 7 not stated; 26 of the 30 (86.7%, 95% interval 70.3 to 94.7%) are not confirmed dead. Every deposited construct carries an engineered substitution; the failure is the inference from substitution to dead enzyme. The tier is retired.

Key columns.

Column	Meaning
verdict	confirmed_reduced, confirmed_inactive, active, not_stated, and the summary rows
n_pairs / denominator	number of pairs and the total (30)
note	interval of the mislabelling rate

Excerpt (first 3 rows).

verdict	n_pairs	denominator	note
confirmed_reduced	15	30	
not_stated	7	30	
confirmed_inactive	4	30	

Table S11. Zymogen-mature evaluation set: all 158 metrics ranked by AUROC (complete Table 5)

File. supplementary/S11_zymogen_21pair_all_metrics.csv · 158 rows × 27 columns · public

Supports. Results 3.1.1; the complete version of Table 5. **Cited in.** Methods 2.2.1; Results 3.1.1.

A row is one metric scored on the 21 pairs (158 in all).

Every metric scored on the 21 pairs that share a ligand: structure-space, PLACER and prediction-based metrics, including whole-protein and sequence-only comparators, ranked by direction-free AUROC of dead against active. The best-of-158 threshold (95th percentile of the best AUROC among 158 unrelated metrics in label-swap simulations) is marked: seven metrics exceed it, all summaries of whole-prediction confidence (pLDDT and per-chain pTM), which are comparators and not main-set metrics.

Key columns.

Column	Meaning
rank / rank_in_table_5	rank among all 158 and rank among the 39 items of Table 5
metric_name / canonical_key / source / input_class	the metric, the distinct metric, structure-space / PLACER / prediction-based, and class
role	main, reference or a supplement role (43 rows carry main because aliases of the 33 main-set metrics are separate rows)
n_pairs / auroc / auroc_ci_lo / auroc_ci_hi	pairs (21) and AUROC with 95% interval over pairs
higher_in / auroc_directed	the form (dead or active) in which the metric is larger, and the directed AUROC
above_best_of_158_threshold / applicability_status	1 if above the threshold; applicability status

The shared block of 12 provenance columns (see Conventions) is also present and is not repeated above.

Excerpt (first 3 rows).

rank	metric_name	role	auroc	auroc_ci_lo	auroc_ci_hi
1	chai_plddt_min	supp_global	0.8821	0.7483	0.9932
2	chai_plddt_best_model	supp_global	0.7914	0.6757	0.9025
3	chai_plddt_mean	reference	0.7914	0.6757	0.9025

Table S12. Reference rows in full, across the activity tests

File. supplementary/S12_reference_rows.csv · 35 rows × 9 columns · D2DCure aggregate (licence of the source data unstated; aggregates only, no per-variant rows)

Supports. Results 3.1 and 3.3; Tables 5, 6, 8 and 9; Methods 2.2 and 2.4. **Cited in.** Methods 2.2.1, 2.4.1; Results 3.1.1, 3.1.2, 3.3.1.

A row is one reference item for one measure (35 in all).

The rows that need no catalytic information, reported in full with intervals: crystallographic resolution, sequence length, net charge and tyrosine fraction (AUROC on the 49 zymogen pairs) and pLDDT and pTM (AUROC on the 28 re-predicted pairs); the five declared BglB baselines (side-chain volume change, negative distance to the catalytic residues, burial, BLOSUM62 and the Rosetta score of the BglB data; signed AUROC on 432 variants); and expected hits per 96-well plate at each keep fraction for tyrosine fraction, net charge and sequence length on the plates.

Key columns.

Column	Meaning
reference_item / metric	the reference row (and the end kept for plate rows)

Column	Meaning
measure	AUROC (direction-free), AUROC (signed, pre-directed) or expected hits per 96-well plate
keep_fraction	fraction of designs kept (plate rows only)
estimate / ci_lo / ci_hi / n	the value, its interval and the number of units

Excerpt (first 3 rows where measure = AUROC (signed, pre-directed)).

reference_item	measure	estimate	ci_lo	ci_hi	n
abs_dvol_A3	AUROC (signed, pre-directed)	0.5513	0.4728	0.6278	432
neg_d_cat	AUROC (signed, pre-directed)	0.7633	0.6872	0.8243	432
neg_rel_sasa	AUROC (signed, pre-directed)	0.6831	0.6058	0.748	432

Notes.

- Contains aggregates of D2DCure data (licence unstated); no per-variant rows.

Table S13. Plated designs: all 128 metrics ranked by the AUROC of active against no active designs (complete Table 6)

File. supplementary/S13_plates_all_metrics_auroc.csv · 128 rows × 29 columns · public

Supports. Results 3.1.2; the complete version of Table 6. **Cited in.** Methods 2.2.2; Results 3.1.2.

A row is one metric scored on the 192 designs (128 in all).

Every metric scored on the 192 plated designs, ranked by the direction-free AUROC of the 16 active designs against the 176 no active designs, with 95% interval over the 136 backbone clusters. The best-of-128 threshold (permuting labels among designs) is marked: ten metrics exceed it. They are two main-set metrics (the intra-residue repulsion, 0.80, and the repulsion of the catalytic residues, 0.78), an alias of the latter, and sequence-composition and whole-protein comparators.

Key columns.

Column	Meaning
rank / rank_in_table_6 / in_table_6	rank among 128, rank among the 36 items of Table 6 and membership of Table 6
metric_name / canonical_key / source / input_class / role	the metric, its source, class and role
n_designs / n_active	192 and 16
auroc / auroc_ci_lo / auroc_ci_hi / auroc_directed / higher_in	AUROC, interval, directed AUROC and the group in which the metric is larger
above_best_of_all_metrics_thresh / applicability_status	1 if above the best-of-128 threshold; applicability status

The shared block of 12 provenance columns (see Conventions) is also present and is not repeated above.

Excerpt (first 3 rows).

rank	metric_name	role	auroc	auroc_ci_lo	auroc_ci_hi
1	rosetta_cat_fa_int ra_rep	main	0.8018	0.6385	0.8929
2	frac_G	supp_global	0.7926	0.6157	0.8698
3	rosetta_ref	excluded_bookkeep ing	0.7876	0.6708	0.8914

Notes.

- Designs are not exchangeable (10 of the 16 active designs lie in two backbone clusters), so the threshold understates the chance range.

Table S14. Substrate swap on all systems: all 33 prediction-based quantities, natural enzymes and de novo designs

File. supplementary/S14_substrate_swap_per_metric.csv · 242 rows × 17 columns · public

Supports. Results 3.2; the all-system analysis behind Table 7. **Cited in.** Methods 2.3; Results 3.2.

A row is one prediction-based quantity under one wrong-ligand condition, for one source (natural enzymes or de novo designs).

The substrate swap before the 55-enzyme evaluation set was fixed: all 59 natural enzymes (cognate value averaged over five seeds) and all 30 de novo designs, for every one of the 33 prediction-based quantities that were scored, including the Chai-1 combined score (a rescaled ipTM) that Table 7 leaves out. For each quantity and each wrong ligand (same enzyme class, different class, decoy; apo for the quantities that are defined without a ligand) it gives the mean under the cognate and under the other condition, the paired change, the change in units of seed noise, and the direction-free AUROC with its interval.

Key columns.

Column	Meaning
source	natural or denovo
metric_name / canonical_key / input_class	the quantity, the distinct metric it belongs to and its class
ame_flavour	crystal = active-site accuracy against the deposited structure; self_design = against the design's own model (never pooled)
in_main_set	1 for the four main-set metrics (each appears several times: mean, maximum and spread over the five predicted models are separate rows)
condition	same_ec, diff_ec, decoy or apo
n_systems	systems with a value (59 natural, 30 de novo)
mean_cognate / mean_condition / mean_delta	mean value under the cognate ligand, under the condition, and the paired difference
median_delta_over_sigma / frac_beyond_1sigma	median change in units of the seed-to-seed standard deviation, and the share of systems that move by more than one
auroc / auroc_lo / auroc_hi	direction-free AUROC of cognate against the condition, 95% interval over systems

Excerpt (first 3 rows where source = natural, condition = diff_ec, in_main_set = 1).

source	metric_name	condition	n_systems	mean_cognate	mean_condition	auroc
natural	ame_crystal_cls_max	diff_ec	58	3.2826	3.6103	0.6564
natural	ame_crystal_rms_d_max	diff_ec	59	1.4469	1.5572	0.5395
natural	ame_crystal_rms_d_mean	diff_ec	59	1.2516	1.2895	0.5306

Notes.

- n is 59 natural enzymes here and 55 in Table 7, whose evaluation set drops four enzymes whose ligand could not be built.
- Per-metric results of this table are descriptive; the ranking and the best-of-30 threshold are in Table 7.

Table S15. Substrate swap stratified by ligand size and charge

File. supplementary/S15_substrate_swap_size_strata.csv · 1,136 rows × 27 columns · public

Supports. Results 3.2; Methods 2.3 (size check, Eq. 10). **Cited in.** Methods 2.3; Results 3.2.

A row is one quantity, source, wrong-ligand condition and stratum of the difference in ligand size or charge, for one analysis variant.

The check of whether ligand size or charge explains the discrimination. For each headline quantity (ipTM, per-chain pTM minimum, the combined score, ligand clearance), comparator (pTM, pLDDT, AME RMSD) and the two baselines (ligand heavy-atom count and formal charge), the cells (an enzyme and one wrong ligand) are stratified by the difference in heavy-atom count (smaller, matched, larger), by the charge difference (same, different) and jointly. For each stratum it gives the paired win fraction, the AUROC of the absolute value, the Spearman correlation of the size difference with the change, intervals over systems and clusters, and whether the effect survives.

Key columns.

Column	Meaning
source / variant	natural or denovo; primary or the variant that excludes ipTM = 0 cells (excl_ipTM0)
metric_name / role	the quantity and headline, comparator or baseline
stratum_family / stratum	all, size (smaller, matched, larger), charge (q_same, q_diff) or joint, and the stratum
condition	same_ec, diff_ec, decoy or the three pooled (pooled_3)
k_cells / N_cells / k_systems / N_systems	cells and systems in the stratum
win_frac / win_lo95 / win_hi95	paired win fraction (cognate higher than wrong ligand) with 95% interval
auroc_abs / spearman_dn_vs_change / survives	AUROC of absolute value, Spearman correlation of size difference with the change, and 1 / 0 / 'n<10, not estimated'

Excerpt (first 3 rows where variant = primary, source = natural, stratum_family = size).

metric_name	stratum	condition	k_cells	N_cells	win_frac
chai_ipTM_mean	smaller	same_ec	18	177	1.0
chai_aggregate_score_mean	smaller	same_ec	18	177	1.0
chai_per_chain_ptm_min_mean	smaller	same_ec	18	177	0.8333

Notes.

- Released as a CSV only for the full 1,136 rows; the excerpt shows the structure.

Table S16. BglB: every analysed quantity ranked by rank correlation with the measured impairment (complete Table 8)

File. supplementary/S16_bglb_all_quantities_ranked.csv · 114 rows × 35 columns · D2DCure aggregate (licence of the source data unstated; aggregates only, no per-variant rows)

Supports. Results 3.3.1; the complete version of Table 8. **Cited in.** Methods 2.4, 2.4.1; Results 3.3.1.

A row is one quantity analysed on the 432 BglB variants (114 in all).

Every quantity analysed on the 432-variant evaluation set: 84 structure-space metrics, 25 prediction-based metrics and the five declared baselines, ranked by the Spearman correlation of the size of its change with the measured impairment, with 95% interval over the 175 positions. It carries the best-of-114 threshold, whether the quantity beats the distance baseline, the correlation of the signed change, and beside them the AUROC of the binary contrast (impaired against wild-type-like variants) with its interval, threshold and beats-the-distance flag. No quantity beats the distance baseline; 34 exceed the best-of-114 threshold.

Key columns.

Column	Meaning
rank / rank_in_table_8 / in_table_8	rank among 114, rank among the 41 items of Table 8 and membership of Table 8
metric_name / canonical_key / source / input_class / role	the quantity, its source (structure-space, prediction-based or baseline), class and role
n_variants	variants with a value (431 or 432; 248 for the Rosetta score of the BglB data)
rho / rho_ci_lo / rho_ci_hi	Spearman correlation and 95% interval over positions
above_best_of_all_quantities_threshold / beats_distance_baseline / diff_vs_distance_baseline_ci_lo	threshold flag, beats-the-distance flag and the lower end of the difference in correlation
rho_signed_delta	correlation of the signed change
auroc / auroc_ci_lo / auroc_ci_hi	AUROC of the binary contrast with 95% interval; the columns that follow (threshold, beats-the-distance flag and difference against the distance baseline) are for the AUROC

The shared block of 12 provenance columns (see Conventions) is also present and is not repeated above.

Excerpt (first 3 rows).

rank	metric_name	role	rho	rho_ci_lo	rho_ci_hi
1	propka_catalytic_p ka_range	main	0.48	0.3474	0.592
2	propka_catalytic_p ka_min	main	0.4622	0.3224	0.5748
3	propka_catalytic_s hft_absmax	main	0.4587	0.319	0.5782

Notes.

- Aggregates of D2DCure data (licence unstated); no per-variant rows.

Table S17. BglB: all 84 structure-space and comparator metrics

File. supplementary/S17_bglb_all_metrics.csv · 84 rows × 29 columns · D2DCure aggregate (licence of the source data unstated; aggregates only, no per-variant rows)
Supports. Results 3.3.1; Methods 2.4.1. **Cited in.** Methods 2.4.1; Results 3.3.1.
A row is one metric analysed on the 432 BglB variants (84 in all).

Spearman correlation, signed, direction-free and magnitude AUROC with intervals, the paired difference against the distance baseline and whether the metric beats all baselines, for every structure-space metric and comparator (36 site-scoped, 28 sequence-only, 15 whole-protein, 5 bookkeeping). None beats all baselines.

Key columns.

Column	Meaning
metric / canonical_key / input_class / in_main_set	the metric, its class and main-set membership
rho / rho_lo / rho_hi	Spearman correlation with 95% interval over positions
auc_signed / auc_dirfree / auc_mag (+_lo, _hi)	AUROC of the binary contrast using the signed change, the direction-free form and the magnitude, with intervals
diff_vs_neg_d_cat_lo / beats_all_baselines	lower end of the difference against the distance baseline; whether the metric beats all five baselines

The shared block of 12 provenance columns (see Conventions) is also present and is not repeated above.

Excerpt (first 3 rows where in_main_set = 1).

metric	in_main_set	rho	rho_lo	rho_hi	beats_all_baselines
catalytic_frac_buried	1	-0.1909	-0.3048	-0.057	0
catalytic_pairwise_dist_max	1	0.0431	-0.1841	0.2709	0
catalytic_pairwise_dist_mean	1	0.09	-0.074	0.2468	0

Notes.

- Aggregates of D2DCure data (licence unstated); no per-variant rows.

Table S18. BglB: sensitivity analyses

File. supplementary/S18_bglb_sensitivities.csv · 11 rows × 22 columns · D2DCure aggregate (licence of the source data unstated; aggregates only, no per-variant rows)
Supports. Results 3.3.1. **Cited in.** Methods 2.4.1; Results 3.3.1.
A row is the main analysis and one of ten sensitivity analyses (11 rows).

The main analysis and ten sensitivity analyses with their sizes, the number of metrics with an interval above 0.5, the number that beat the distance baseline and the AUROC of the distance baseline. The analyses are: the main analysis (432 variants); variants near to and distal from the catalytic residues (144 and 288); variants with a measured stability (292); each institution left out in turn (FIU, MiraCosta, UC Davis); other thresholds for the impaired label (1 and 3 standard deviations, with 173 and 139 impaired variants); responding metrics only (70 metrics analysed); and without the variants at catalytic positions (431 variants, one position removed). In every analysis no metric beats the distance baseline.

Key columns.

Column	Meaning
analysis	the analysis
n_variants / n_positions / n_impaired / n_wtlike	size of the analysis
n_metrics_analysed / n_auc_mag_ci_above_half	metrics analysed and metrics whose AUROC interval lies above 0.5
n_beats_neg_d_cat / n_beats_all_baselines	metrics that beat the distance baseline and all baselines
auc_neg_d_cat_baseline	AUROC of the distance baseline

The shared block of 12 provenance columns (see Conventions) is also present and is not repeated above.

Excerpt (first 3 rows).

analysis	n_variants	n_positions	n_impaired	n_wtlike	n_beats_neg_d_cat
main analysis	432	175	166	254	0
variants distal from the catalyti...	288	136	71	207	0
variants near the catalytic resid...	144	39	95	47	0

Notes.

- Aggregates of D2DCure data (licence unstated); no per-variant rows.

Table S19. BglB: prediction-based metrics (Chai-1 with the assay substrate)

File. supplementary/S19_bglb_prediction_based_all_metrics.csv · 28 rows × 29 columns · D2DCure aggregate (licence of the source data unstated; aggregates only, no per-variant rows)

Supports. Results 3.3.1; Methods 2.4.1. **Cited in.** Methods 2.4.1; Results 3.3.1.

A row is one prediction-based metric analysed on the 432 BglB variants (28 in all).

The same columns as S17 for the metrics computed from Chai-1 predictions of each variant with the assay substrate (pNPG, no metal); active-site accuracy is measured against the wild-type crystal, which carries a covalent glucosyl intermediate. Twelve rows are main-set metrics; the largest correlation is 0.17.

Key columns.

Column	Meaning
metric / canonical_key / input_class / in_main_set	the metric, its class and main-set membership
rho / rho_lo / rho_hi	Spearman correlation with 95% interval over positions
auc_signed / auc_dirfree / auc_mag (+_lo, _hi)	AUROC of the binary contrast
beats_all_baselines	whether the metric beats all five baselines

The shared block of 12 provenance columns (see Conventions) is also present and is not repeated above.

Excerpt (first 3 rows where in_main_set = 1).

metric	in_main_set	rho	rho_lo	rho_hi
ame_crystal_clash_max	1	0.0828	-0.0534	0.2162
ame_crystal_rmsd_max	1	0.0632	-0.0622	0.1849
ame_crystal_rmsd_mean	1	0.1213	-0.0079	0.259

Notes.

- Aggregates of D2DCure data (licence unstated); no per-variant rows.

Table S20. Plated designs: hits per 96-well plate for every metric (filter-style analysis)

File. supplementary/S20_plates_hits_per_plate_ranked.csv · 128 rows × 37 columns · public

Supports. Results 3.3.2; Methods 2.4.2 (Eqs. 16 and 17). **Cited in.** Methods 2.4.2; Results 3.3.2.

A row is one metric scored on the 192 designs (128 in all).

The filter-style analysis, for every metric: the expected hits per 96-well plate and the actives among the 48 designs kept when the quarter of designs that the metric ranks highest, or lowest, fills the plate (8.0 hits when unfiltered), each with a 90% interval over designs, ranked by the better end. 30 metrics exceed the random-filter band (14.0 hits).

Key columns.

Column	Meaning
rank / rank_among_ranked_items / ranked_item	rank among 128, rank among the 36 items of Table 9 and whether the metric is one of them
metric_name / canonical_key / source / input_class / role	the metric and its class
n_designs / n_active / n_kept_highest / n_kept_lowest	192, 16 and the number kept (fewer than 48 when designs tie at the cut)
hits_highest_quarter / hits_highest_ci_lo90 / hits_highest_ci_hi90 / actives_in_highest_quarter	hits per plate for the highest quarter, 90% interval and actives kept
hits_lowest_quarter / hits_lowest_ci_lo90 / hits_lowest_ci_hi90 / actives_in_lowest_quarter	the same for the lowest quarter
better_end / best_hits_per_plate / above_random_band	the better end, its hits per plate and whether it exceeds the random band
note	for example a tie at the cut

The shared block of 12 provenance columns (see Conventions) is also present and is not repeated above.

Excerpt (first 3 rows).

rank	metric_name	role	better_end	best_hits_per_plate	above_random_band
1	frac_Y	reference	highest	24.0	1
2	frac_G	supp_global	lowest	22.0	1
3	rosetta_cat_fa_int ra_rep	main	lowest	22.0	1

Notes.

- Descriptive only: no hypothesis test.

Table S21. Plated designs: every metric ranked by correlation with kcat/KM among the 16 designs that have a value (complete Table 9)

File. supplementary/S21_plates_all_metrics_ranked.csv · 106 rows × 29 columns · public

Supports. Results 3.3.2; the complete version of Table 9. **Cited in.** Methods 2.4.2; Results 3.3.2.

A row is one metric with a rank correlation on the 16 designs with a reported kcat/KM (106 in all).

Every metric that has a correlation, ranked by the size of the Spearman correlation with log₁₀ kcat/KM, with a 90% interval over designs, whether the interval excludes 0 (33 do), the direction, and the correlation with the sequence length partialled out and whether it survives that control.

Key columns.

Column	Meaning
rank / rank_in_table_9 / in_table_9	rank among 106, rank among the 36 items of Table 9 and membership of Table 9
metric_name / canonical_key / source / input_class / role	the metric and its class
n_designs / rho / rho_ci_lo90 / rho_ci_hi90	16 designs, Spearman correlation and 90% interval over designs
interval_excludes_zero / higher_value_means	whether the interval excludes 0 and whether a larger value means higher or lower activity
rho_given_length / survives_length_control	correlation with the sequence length partialled out of both sides, and whether it survives

The shared block of 12 provenance columns (see Conventions) is also present and is not repeated above.

Excerpt (first 3 rows).

rank	metric_name	rho	rho_ci_lo90	rho_ci_hi90	higher_value_means
1	frac_I	-0.7821	-0.9494	-0.5042	lower activity
2	frac_A	0.776	0.4627	0.9348	higher activity
3	rosetta_omega	-0.7756	-0.8904	-0.5	lower activity

Notes.

- Descriptive only; about 10% of unrelated metrics are expected to have an interval that excludes 0.

Table S22. Plated designs: ordering among the 16 active designs, structure-space metrics and comparators

File. supplementary/S22_plates_ordering.csv · 81 rows × 23 columns · public

Supports. Results 3.3.2; Methods 2.4.2 (Eq. 15). **Cited in.** Methods 2.4.2; Results 3.3.2.

A row is one metric of the structure-space panel or a comparator (81 in all).

The Spearman correlation with log kcat/KM among the 16 active designs with and without the sequence length partialled out, with 90% intervals and whether the interval excludes 0 (24 do).

Key columns.

Column	Meaning
metric_name / canonical_key / input_class / in_main_set	the metric and its class
spearman_rho / rho_lo90 / rho_hi90 / excludes_zero	correlation, 90% interval and whether it excludes 0
rho_given_length / survives_length_control	correlation with length partialled out and whether it survives

The shared block of 12 provenance columns (see Conventions) is also present and is not repeated above.

Excerpt (first 3 rows where in_main_set = 1).

metric_name	in_main_set	spearman_rho	rho_lo90	rho_hi90	rho_given_length
catalytic_polar_contacts	1	0.6711	0.4115	0.8501	0.5183
rosetta_catalytic_vdw_net	1	-0.6299	-0.8777	-0.2414	-0.4743
rosetta_cat_fa_sol	1	0.5651	0.1543	0.8487	0.538

Table S23. Plated designs: enrichment at every keep fraction, structure-space metrics

File. supplementary/S23_plates_enrichment.csv · 816 rows × 27 columns · public

Supports. Results 3.3.2; Methods 2.4.2. **Cited in.** Methods 2.4.2; Results 3.3.2.

A row is one metric, one direction (keep the highest or the lowest values) and one keep fraction (816 rows).

Enrichment on the 192 plated designs for the 102 metrics of the structure-space panel and its comparators, in both directions and at keep fractions of 5, 10, 25 and 50%: the number kept, the actives kept and the expected hits per 96-well plate with a 90% interval against the unfiltered 8.0.

Key columns.

Column	Meaning
metric / canonical_key / input_class / in_main_set	the metric
direction / keep_fraction / n_kept	keep the highest or the lowest values; the fraction and the number of designs kept
n_active_kept / n_active_total	actives among those kept, and 16
expected_hits_per_plate / hits_lo90 / hits_hi90	expected hits per plate with 90% interval
baseline_hits_per_plate / enrichment_vs_baseline	8.0 and the ratio to it

The shared block of 12 provenance columns (see Conventions) is also present and is not repeated above.

Excerpt (first 3 rows where in_main_set = 1, keep_fraction = 0.25).

metric	direction	keep_fraction	n_kept	n_active_kept	expected_hits_per_plate
catalytic_frac_buried	low	0.25	48	2	4.0
catalytic_frac_buried	high	0.25	48	1	2.0
catalytic_pairwise_dist_max	low	0.25	48	3	6.0

Notes.

- Descriptive only. Released as a CSV only for the full 816 rows.

Table S24. Plated designs: enrichment for prediction-based metrics

File. supplementary/S24_plates_prediction_based_enrichment.csv · 232 rows × 27 columns · public

Supports. Results 3.3.2; Methods 2.4.2. **Cited in.** Methods 2.4.2; Results 3.3.2.

A row is one prediction-based metric, one direction and one keep fraction (232 rows).

The same as S23 for the 29 prediction-based metrics (Chai-1 with the transition-state analogue and zinc; active-site accuracy against the design's own model).

Key columns.

Column	Meaning
metric / canonical_key / input_class / in_main_set	the metric
direction / keep_fraction / n_kept / n_active_kept	as in S23
expected_hits_per_plate / hits_lo90 / hits_hi90 / enrichment_vs_baseline	expected hits per plate, 90% interval and ratio to 8.0

The shared block of 12 provenance columns (see Conventions) is also present and is not repeated above.

Excerpt (first 3 rows where keep_fraction = 0.25, direction = high).

metric	direction	keep_fraction	n_active_kept	expected_hits_per_plate
chai_plddt_mean	high	0.25	8	16.0
chai_ptm_mean	high	0.25	9	18.0
frac_Y	high	0.25	12	24.0

Notes.

- Descriptive only (16 active designs).

Table S25. Sanity floors by class of metric

File. supplementary/S25_sanity_floor_by_input_class.csv · 53 rows × 8 columns · public

Supports. Methods 2.5 (detection floor); Results 3.4.1. **Cited in.** Methods 2.5; Results 3.4.1.

A row is one sanity-floor condition on one data set and one class of metric (53 rows).

The sanity-floor conditions (all catalytic residues replaced by Gly, scrambled sequence on the native backbone, unrelated protein, ligand removed on the pilot set) for the 143-enzyme set, the shortened-motif set, the pilot set and the de novo set: how many metrics of each class were scored and how many responded. One row reconciles the detection floor with the Gly step: the responses are identical on 98 of 98 metrics (the same change scored twice).

Key columns.

Column	Meaning
dataset / condition / input_class	the data set, the condition and the class of metric
n_metrics / n_scored / n_responds / n_unscored	metrics in the class, scored, responding and not scored
note	remarks, including the reconciliation

Excerpt (first 3 rows where dataset = main set (143), condition = cat_to_gly).

dataset	condition	input_class	n_metrics	n_scored	n_responds
main set (143)	cat_to_gly	bookkeeping	20	20	8
main set (143)	cat_to_gly	seq_only	29	29	10
main set (143)	cat_to_gly	struct_site	34	26	25

Table S26. Control matching: the loosest tier needed

File. supplementary/S26_control_match_tiers.csv · 81 rows × 6 columns · public

Supports. Methods 2.5; Results 3.4.1. **Cited in.** Methods 2.5; Results 3.4.1.

A row is one analysis, lesion test and matching tier (81 rows).

For each analysis and lesion test, how many (enzyme, step) entries needed each loosest matching tier: 0 = exact match, tiers 3 and above drop packing matching, tier 5 also relaxes burial (main set only), -1 = no control found; and the share of the lesion test.

Key columns.

Column	Meaning
analysis_set / lesion_test	the analysis (data set and controls) and the lesion test (catalytic lesion, second-shell lesion or the retired deformation experiment)
worst_tier / n_entries / share_of_lesion_test	loosest tier used, number of entries and their share of the lesion test
note	definition of the tiers

Excerpt (first 3 rows where analysis_set = main set, packing-matched controls, lesion_test = catalytic lesion).

analysis_set	lesion_test	worst_tier	n_entries	share_of_lesion_test
main set, packing-matched controls	catalytic lesion	-1.0	8	0.0035
main set, packing-matched controls	catalytic lesion	0.0	226	0.0997
main set, packing-matched controls	catalytic lesion	2.0	496	0.2189

Table S27. Control matching: balance of residues changed in the two arms

File. supplementary/S27_control_arm_balance.csv · 15 rows × 9 columns · public

Supports. Methods 2.5; Results 3.4.1. **Cited in.** Methods 2.5; Results 3.4.1.

A row is one analysis and lesion test (15 rows).

The share of (enzyme, step) pairs in which the control arm changed fewer or more residues than the catalytic arm, the median and mean ratio, and the number with no control. The comparison of medians used elsewhere cannot see this imbalance.

Key columns.

Column	Meaning
analysis_set / lesion_test / pairs	the analysis, the lesion test and the number of pairs
control_fewer_share / control_more_share	shares with fewer or more residues changed in the control arm
median_ratio / mean_ratio / catalytic_only_no_control	ratio of residues changed and pairs with no control

Excerpt (first 3 rows).

analysis_set	lesion_test	pairs	control_fewer_share	control_more_share	median_ratio
pilot set	catalytic lesion	212	0.1981	0.0	1.0
pilot set	second-shell lesion	159	0.0	0.0	1.0
pilot set	deformation	205	0.0	0.0	1.0

Table S28. Catalytic lesion: every metric scored (Table 10 plus supplementary comparators and per-step flags)

File. supplementary/S28_catalytic_lesion_all_metrics_ranked.csv · 43 rows × 73 columns · public

Supports. Results 3.4.1; the complete version of Table 10. **Cited in.** Methods 2.5.1; Results 3.4.1.

A row is one metric scored on the catalytic-lesion ladder (43 in all).

Table 10 with the 8 supplementary comparators and every flag that decides whether a step counts. Metrics are grouped by the lesion step at which they are first detected and ranked by the mean of the rank statistic R over the counted steps. For each step (isosteric 15.3, non-isosteric 36.4, Ala 53.2, Gly 80.5 A3) it gives R with interval, the number of informative proteins and the flags that the metric responds, that the control response is above its noise and that the metric is specific; the number of steps counted; the first step detected and the first step at which the metric is specific (the onset of specificity); the mean R of the 143 main enzymes alone; and, for the prediction-based metrics, the distance-matched R. Structure-space metrics: 195 pooled natural enzymes (143 main + 52 pilot); prediction-based: 59 enzymes, 286 pairs.

Key columns.

Column	Meaning
detection_group / rank_in_group / rank / rank_in_table_10	group by first detected step, rank in the group, overall rank and rank among the 35 items of Table 10
metric_name / canonical_key / source / input_class / role / in_ranked_set	the metric, class, role and whether it is one of the 35 ranked items
n_steps / n_steps_interpretable	steps with a value and steps counted
r_mean / r_lo / r_hi	mean R over counted steps with 95% interval over enzyme sub-subclasses
r_ / _lo / hi / n	R, interval and informative proteins for isosteric, non_isosteric, ala and gly
responds_ / control_above_noise_ / specific_	the per-step flags
first_step_detected / first_step_specific	onsets of detection and of specificity
r_matched / n_pairs_matched / n_systems_matched	distance-matched R and its size (prediction-based metrics)
r_mean_primary_only / above_best_of_all_quantities_threshold	mean R of the 143 main enzymes and the best-of-N flag

The shared block of 12 provenance columns (see Conventions) is also present and is not repeated above.

Excerpt (first 3 rows).

detection_group	rank_in_group	metric_name	r_mean	r_lo	r_hi
isosteric	1	chai_per_chain_ptm_min_mean	0.7524	0.6768	0.8238
isosteric	2	propka_catalytic_pka_range	0.6834	0.5759	0.7806
isosteric	3	chai_iptm_mean	0.6732	0.5991	0.7463

Notes.

- Descriptive of a pooled set. [S29](#) and [S30](#) hold the specificity-ratio analysis.

Table S29. Structure-space specificity by metric family and lesion step, with the panel statistic and sensitivity sets

File. supplementary/S29_specificity_by_family_and_step.csv · 20 rows × 27 columns · public

Supports. Results 3.4.1 and 3.4.2. **Cited in.** Methods 2.5.1; Results 3.4.1, 3.4.2.

A row is one family of metrics at one lesion step or dose (20 rows).

The number of eligible, responding and specific structure-space metrics (as distinct metrics) by family (site-scoped Rosetta energy, catalytic geometry, catalytic pKa, all main-set metrics) at each step of the catalytic lesion and each second-shell dose, the geometric-mean specificity ratio of the main-set panel (isosteric, 1.075 [0.958, 1.178]), and the counts in four sensitivity sets (shortened motif, no packing matching or count gate, pilot set, de novo designs). At the isosteric step 23 of 29 metrics respond and none is specific; specificity appears for 5 metrics at the non-isosteric step and for 8 at the Ala and Gly steps.

Key columns.

Column	Meaning
family / lesion_test / level / dose_value / dose_unit	family, lesion test (catalytic lesion or second-shell lesion), step or dose and its size
n_eligible_keys / n_responds_keys / n_specific_keys / n_cells	counts as distinct metrics, and cells
sr_geomean_ci90	geometric-mean specificity ratio of the panel with 90% interval
sens_shortened_motif / sens_no_packing_match / sens_pilot_set / sens_de_novo_set	specific / eligible in the shortened-motif, no-packing-match, pilot and de novo sets

The shared block of 12 provenance columns (see Conventions) is also present and is not repeated above.

Excerpt (first 3 rows where lesion_test = catalytic lesion).

family	level	n_eligible_keys	n_responds_keys	n_specific_keys	sens_shortened_motif
Rosetta energy (site-scoped)	isosteric	15	13	0	0 of 15
Catalytic geometry (site-scoped)	isosteric	8	5	0	0 of 8
Catalytic pKa	isosteric	6	5	0	

Table S30. Re-predicted catalytic lesion: the prediction-based metrics under three successive controls and the pre-specified rule

File. supplementary/S30_predictor_ladder_three_controls.csv · 6 rows × 23 columns · public

Supports. Results 3.4.1; Methods 2.5.1 (Eq. 24). **Cited in.** Methods 2.5.1; Results 3.4.1.

A row is one prediction-based metric or reference row (6 rows).

Specificity of the prediction-based metrics on the natural arm of the re-predicted ladder (59 enzymes, 286 pairs): the specificity ratio on the isosteric step with the original control, in the distance-matched subset (68 pairs, 21 systems) and after regression adjustment on all steps (with the unadjusted ratio on the same pairs), and the label of the rule specified before the analysis. The ratio of the interface terms falls once the control is matched or adjusted on the distance to the ligand (ipTM: 1.96 with the original control, 0.47 in the distance-matched subset and 1.18 after adjustment); no interface term meets the rule.

Key columns.

Column	Meaning
metric / role	the metric and main or reference
n_pairs_isosteric / n_systems_isosteric	pairs and systems at the isosteric step
sr_mean_isosteric_original_control	specificity ratio with the original control
distance_matched_k_of_N / sr_distance_matched	size of the distance-matched subset and the ratio in it
unadjusted_gmr_all_steps / adjusted_gmr_all_steps	geometric-mean ratio over all steps, unadjusted and adjusted for distance and burial
rule_label	the label from the pre-specified rule (lower 90% bound above 1.2 with at least 30 pairs in 15 systems)

The shared block of 12 provenance columns (see Conventions) is also present and is not repeated above.

Excerpt (first 3 rows).

metric	sr_mean_isosteric_original_control	sr_distance_matched	adjusted_gmr_all_steps	rule_label
ame_crystal_clash_max	1.84 [1.31, 2.69]	1.05 [0.79, 1.39]	1.18 [0.95, 1.46]	testable; not specific (lower 90%...
ame_crystal_rmsd_min	2.67 [1.02, 4.74]	0.98 [0.68, 1.21]	1.90 [1.40, 2.53]	knockon_displacement_no_specifici...
chai_ipTM_mean	1.96 [1.44, 2.71]	0.47 [0.24, 1.04]	1.18 [0.85, 1.60]	testable; not specific (lower 90%...

Table S31. Sensitivity of the specificity-ratio counts to motif size, packing matching and count gating

File. supplementary/S31_motif_packing_sensitivity.csv · 20 rows × 7 columns · public

Supports. Results 3.4.1. **Cited in.** Methods 2.5.1; Results 3.4.1.

A row is one analysis and one lesion step (20 rows).

Counts of eligible, responding and specific metrics at each step for five analyses: the pilot set, the main set without packing matching or count gating, the main set with packing-matched controls (primary), the main set with the motif shortened to at most 3 residues, and the 30 de novo designs with a created-site control.

Key columns.

Column	Meaning
reader_name / motif_and_controls	the analysis and how its motif and controls are defined
level	lesion step
n_eligible_keys / n_responds_keys / n_specific_keys / n_cells	counts as distinct metrics, and cells

Excerpt (first 3 rows).

reader_name	level	n_eligible_keys	n_responds_keys	n_specific_keys
pilot set	isosteric	23	12	0
pilot set	non_isosteric	23	12	0
pilot set	ala	23	14	0

Table S32. Re-predicted ladder: seed-noise context

File. supplementary/S32_replicate_noise_context.csv · 56 rows × 12 columns · public

Supports. Methods 2.5.1; Results 3.4.1. **Cited in.** Methods 2.5.1; Results 3.4.1.

A row is one prediction-based metric in one arm and subset (56 rows).

Noise context from the second-seed replicate on 30 natural pairs: the median absolute change in each arm against the seed noise of the metric, for the nine prediction-based metrics, on all pairs and on the original matched subset.

Key columns.

Column	Meaning
source / metric_name / subset / arm	natural or de novo, the metric, all_pairs or M0_matched, and catalytic or control
k_pairs / N_pairs / k_systems / N_systems	pairs and systems
median_abs_delta / median_sigma_native / frac_abs_delta_gt_sigma_native	median absolute change, the metric's seed noise and the share of pairs whose change exceeds it

Excerpt (first 3 rows where source = natural, subset = all_pairs).

metric_name	subset	arm	median_abs_delta	median_sigma_native	frac_abs_delta_gt_sigma_native
chai_iptm_mean	all_pairs	catalytic	0.00666	0.00154	0.8292
chai_iptm_mean	all_pairs	control	0.00254	0.00154	0.6797
chai_aggregate_score_mean	all_pairs	catalytic	0.00556	0.00127	0.8392

Table S33. Specificity ratio of every eligible metric at every step

File. supplementary/S33_eligible_metrics.csv · 256 rows × 29 columns · public

Supports. Methods 2.5.1 (Eq. 22); Results 3.4.1 and 3.4.2. **Cited in.** Methods 2.5.1; Results 3.4.1, 3.4.2.

A row is one eligible (site-scoped) metric at one lesion step or dose (256 rows).

The median change at the catalytic lesion and its interval, the specificity ratio with 95% interval, whether the metric responds and whether it is specific, and the verdict: specific, non_specific, blind (does not respond), invariant (does not change) or below the floor of five units. Steps of the catalytic lesion are isosteric, non_isosteric, ala, gly; second-shell doses are second_shell_1, _2, _4.

Key columns.

Column	Meaning
canonical_key / metric_name / input_class / in_main_set	the metric (aliases collapse in canonical_key)
lesion_test / level / n_systems	catalytic lesion or second-shell lesion, the step or dose and the number of enzymes
delta_median / delta_ci_lo / delta_ci_hi	median change at the catalytic lesion with interval
sr_median / sr_ci_lo / sr_ci_hi	specificity ratio with 95% interval
responds / specific / verdict / applicability_status	the flags, the verdict and the applicability status

The shared block of 12 provenance columns (see Conventions) is also present and is not repeated above.

Excerpt (first 3 rows where in_main_set = 1, level = ala).

canonical_key	lesion_test	level	n_systems	sr_median	verdict
catalytic_frac_buried	catalytic lesion	ala	143	1.9091	specific
catalytic_pairwise_dist_max	catalytic lesion	ala	124	0.0	blind
catalytic_pairwise_dist_mean	catalytic lesion	ala	124	0.7262	non_specific

Table S34. Specificity ratio of whole-protein and sequence-only comparators at every step

File. supplementary/S34_global_comparators.csv · 308 rows × 29 columns · public

Supports. Methods 2.5.1; Results 3.4. **Cited in.** Methods 2.5.1; Results 3.4.1, 3.4.2.

A row is one comparator metric at one lesion step or dose (308 rows).

The same cells as S33 for the whole-protein and sequence-only metrics, which are not in the main set. They show that composition metrics have a ratio of 1 by construction and that whole-protein energies move with the lesion.

Key columns.

Column	Meaning
canonical_key / metric_name / input_class	the comparator
lesion_test / level / n_systems	as in S33
delta_, sr_, responds, specific, verdict	as in S33

The shared block of 12 provenance columns (see Conventions) is also present and is not repeated above.

Excerpt (first 3 rows where level = ala).

canonical_key	lesion_test	level	n_systems	sr_median	verdict
frac_A	catalytic lesion	ala	143	1.1744	non_specific
frac_C	catalytic lesion	ala	143	nan	blind
frac_D	catalytic lesion	ala	143	0.9842	non_specific

Table S35. The specific cells that survive the controls

File. supplementary/S35_survivors_by_status.csv · 97 rows × 10 columns · public

Supports. Methods 2.5.1; Results 3.4.1. **Cited in.** Methods 2.5.1; Results 3.4.1.

A row is one metric at one step in one analysis (97 rows: 74 natural, 23 de novo).

The cells that are specific under the control of their analysis, with the specificity ratio, the population (natural or de novo) and the applicability status of the metric. Of the 97, 69 are of metrics with status OK, 27 of whole-protein comparators (GLOBAL) and 1 of a metric with too little coverage.

Key columns.

Column	Meaning
analysis_set / lesion_test / level	the analysis, the lesion test and the step
metric_name / canonical_key / input_class / in_main_set	the metric
sr_text	specificity ratio with interval, as text
applicability_status / population	applicability status and natural or de novo

Excerpt (first 3 rows where analysis_set = main set, packing-matched controls, in_main_set = 1).

analysis_set	lesion_test	level	metric_name	sr_text	population
main set, packing-matched controls	catalytic lesion	ala	catalytic_frac_buried	1.91 [1.4, 2.72]	natural
main set, packing-matched controls	catalytic lesion	ala	catalytic_radius_of_gyration	1.76 [1.43, 2.19]	natural
main set, packing-matched controls	catalytic lesion	ala	catalytic_rel_sasa_mean	2.4 [1.93, 3.03]	natural

Table S36. Rank statistic R per lesion step on the 143 main enzymes

File. supplementary/S36_ranking_statistic_by_lesion_step.csv · 313 rows × 14 columns · public

Supports. Methods 2.5.1 (Eq. 19). **Cited in.** Methods 2.5.1; Results 3.4.1.

A row is one metric at one step of the catalytic lesion (313 rows).

The rank statistic R (probability that a metric changes more at the catalytic lesion than at the matched control) for the 143 main enzymes alone, with 90% intervals, for 83 distinct metrics including comparators. This is the per-step form of R on the main enzymes; Table 10 and S28 report the pooled analysis on 195 enzymes with 95% intervals and the counted-step rule, so the values for the same metric differ.

Key columns.

Column	Meaning
metric_name / canonical_key / input_class / in_main_set	the metric
lesion_step / volume_A3	lesion step and its median side-chain volume change
R / R_lo90 / R_hi90 / n_systems / n_clusters	R with 90% interval, enzymes and enzyme sub-subclasses
responds / interpretable	whether the metric responds and whether R is interpretable
applicability_status	applicability status

Excerpt (first 3 rows where in_main_set = 1, lesion_step = ala).

metric_name	lesion_step	R	R_lo90	R_hi90	n_systems
catalytic_frac_buried	ala	0.7557	0.686	0.8189	131
catalytic_pairwise_dist_max	ala	0.203	0.1522	0.2574	133
catalytic_pairwise_dist_mean	ala	0.3262	0.2553	0.4	141

Table S37. Re-predicted ladder: distance of the substituted and control residues to the ligand

File. supplementary/S37_ladder_pair_distances.csv · 312 rows × 34 columns · public

Supports. Methods 2.5.1; Results 3.4.1. **Cited in.** Methods 2.5.1; Results 3.4.1.

A row is one catalytic/control pair of the ladder (312 pairs: 286 natural, 26 de novo).

For every pair the residue types, the burial of each residue, the distance of the substituted and of the control residue to the ligand (in the reference structure and in the prediction), whether the control is adjacent to a catalytic residue, and whether the pair falls in each of the calipers used in [S40](#).

Key columns.

Column	Meaning
source / base_system / level	natural or denovo, the system and the lesion step
wt / new_aa / cat_position / ctl_position	residue substituted and its position in the catalytic and control arms
cat_rel_sasa / ctl_rel_sasa / abs_diff_rel_sasa	relative accessible area of the two residues and the difference
d_lig_cat_ref / d_lig_ctl_ref / abs_diff_d_lig_ref	distance to the ligand in the reference structure (A)
d_lig_cat_chai / d_lig_ctl_chai	distance in the Chai-1 prediction
control_adjacent_to_catalytic / ligand_status	adjacency to a catalytic residue; whether the ligand could be located
in_caliper_*	membership of the primary caliper and of sensitivity calipers

Excerpt (first 3 rows where source = natural, level = isosteric, ligand_status = ok).

base_system	level	wt	new_aa	d_lig_cat_ref	d_lig_ctl_ref	abs_diff_rel_sasa
M0024_1nzy_a2	isosteric	F	L	3.433	24.224	0.0031
M0024_1nzy_a2	isosteric	F	Y	3.433	10.504	0.0008
M0050_1dbt_a1	isosteric	D	N	3.703	17.378	0.0002

Notes.

- Released as a CSV only for the full 312 rows.

Table S38. Re-predicted ladder: candidates for a new distance-matched control

File. supplementary/S38_new_distance_matched_controls.csv · 291 rows × 17 columns · public

Supports. Methods 2.5.1; Results 3.4.1. **Cited in.** Methods 2.5.1; Results 3.4.1.

A row is one candidate control for one pair that had no control within the caliper (291 rows).

The candidate controls for the pairs that had none in the caliper, which were selected (54 pairs, 53 new predictions), their distances and burial, and the reason when none was selected (no candidate in the caliper for 213 pairs; no reference distance for 20; no ligand in the baseline prediction for 4).

Key columns.

Column	Meaning
base_system / source / level	system, natural or de novo, step
wt / new_aa / cat_position / orig_ctl_position / new_ctl_position	the substitution and the original and new control positions
d_lig_cat_ref / d_lig_ctl_new / abs_diff_d	distances to the ligand and their difference
cat_rel_sasa / ctl_rel_sasa_new / abs_diff_rel	burial and its difference
n_candidates / selected / reason	number of candidates, 1 if selected, and the reason when not

Excerpt (first 3 rows where selected = 1).

base_system	level	new_ctl_position	d_lig_cat_ref	d_lig_ctl_new	selected	reason
DN_campaign1_E4	ala	30	2.16	2.6964	1	selected

base_system	level	new_ctl_positio n	d_lig_cat_ref	d_lig_ctl_new	selected	reason
M0024_1nzy_a2	ala	82	3.433	4.512	1	selected
M0050_1dbt_a1	isosteric	193	3.703	5.0744	1	selected

Table S39. Re-predicted ladder: specificity ratio in strata of the distance to the ligand

File. supplementary/S39_ladder_distance_strata.csv · 203 rows × 27 columns · public

Supports. Methods 2.5.1; Results 3.4.1. **Cited in.** Methods 2.5.1; Results 3.4.1.

A row is one metric in one distance stratum, for one source (203 rows).

The specificity ratio of each metric in the near, mid and far third of the pairs by the distance of the substituted residue to the ligand (edges 2.73 and 3.66 Å), and on all pairs, with the median distances of the two arms.

Key columns.

Column	Meaning
source / metric_name / role / applicability_status / headline	the metric, interface or global, applicability status and whether it is a headline row
stratum / edge_lo / edge_hi / k_pairs / k_systems	the stratum and its size
SR_mean (+ lo/hi 95 and 90) / SR_median (+ lo/hi 95)	specificity ratio of the mean and of the median change with intervals
mean_abs_delta_cat / mean_abs_delta_ctl	mean absolute change in the two arms
median_d_lig_cat / median_d_lig_ctl / median_abs_diff_d_lig	median distances to the ligand

Excerpt (first 3 rows where source = natural, metric_name = chai_iptm_mean).

metric_name	stratum	k_pairs	k_systems	SR_mean	SR_mean_lo95	SR_mean_hi95
chai_iptm_mean	all_pairs_original_gate	286	59	2.0359	1.4653	2.8031
chai_iptm_mean	all_pairs	281	58	2.0359	1.4532	2.833
chai_iptm_mean	near_third	89	27	2.4894	1.2129	4.7288

Table S40. Re-predicted ladder: caliper-matched subsets and the rule labels

File. supplementary/S40_ladder_matched_subsets.csv · 348 rows × 25 columns · public

Supports. Methods 2.5.1; Results 3.4.1 (68 to 69 pairs in 21 to 22 systems). **Cited in.** Methods 2.5.1; Results 3.4.1.

A row is one metric in one matched subset for one source (348 rows).

The specificity ratio in caliper-matched subsets: original_controls (original controls within the caliper), original_plus_new_controls (with the new controls added, the subset of Table 10) and the sensitivity sets distance_in_prediction, distance_to_organic_atoms, caliper_1A and caliper_3A that use other distance definitions and calipers, with the label from the pre-specified rule (lower 90% bound above 1.2 with at least 30 pairs in 15 systems).

Key columns.

Column	Meaning
source / metric_name / role / applicability_status	the metric and its status
set_id / control	the matched subset and original or original+new controls
k_pairs / k_systems / N_pairs / N_systems	subset size and total
SR_mean (+ lo/hi) / SR_median (+ lo/hi)	specificity ratio with intervals

Column	Meaning
verdict / rule_applies	the label (for example not_testable (coverage), global_comparator_no_verdict) and whether the pre-specified rule applies

Excerpt (first 3 rows where source = natural, metric_name = chai_ipm_mean).

metric_name	set_id	k_pairs	k_systems	SR_mean	verdict
chai_ipm_mean	original_controls	17	16	0.3831	not_testable (coverage)
chai_ipm_mean	distance_in_prediction	23	22	0.4155	sensitivity: not_testable (coverage)
chai_ipm_mean	distance_to_organics_atoms	16	15	0.369	sensitivity: not_testable (coverage)

Table S41. Re-predicted ladder: regression-adjusted ratios

File. supplementary/S41_ladder_regression_adjusted.csv · 116 rows × 28 columns · public

Supports. Methods 2.5.1 (Eq. 24); Results 3.4.1. **Cited in.** Methods 2.5.1; Results 3.4.1.

A row is one metric with the original or the extended set of controls, for one source (116 rows).

The ratio of catalytic to control change adjusted for the distance to the ligand and the burial, $\log(|\Delta| + \epsilon) \sim \text{arm} + d_{\text{lig}} + \text{relSASA}$, beside the unadjusted ratio on the same pairs, and the regression coefficients.

Key columns.

Column	Meaning
source / metric_name / role / applicability_status	the metric
model / control	the regression and original_controls or plus_new_controls
k_pairs / k_systems / n_rows / eps	size and the small constant that keeps zero changes finite
adj_GMR_cat_over_ctl (+ lo/hi95)	adjusted geometric-mean ratio
unadj_GMR_cat_over_ctl (+ lo/hi95)	unadjusted ratio on the same pairs
beta_d_lig_per_A / beta_relSASA (+ lo/hi95)	coefficients per A of distance and per unit of relative accessible area

Excerpt (first 3 rows where source = natural, metric_name = chai_ipm_mean).

metric_name	control	k_pairs	adj_GMR_cat_over_ctl	unadj_GMR_cat_over_ctl
chai_ipm_mean	original_controls	261	1.1785	2.0198
chai_ipm_mean	plus_new_controls	261	1.0882	1.7922

Table S42. Panel equivalence statistic: earlier whole panel against the main-set metrics

File. supplementary/S42_equivalence_earlier_vs_main_panel.csv · 6 rows × 10 columns · public

Supports. Results 3.4.1; Methods 2.5.1 (equivalence to 1). **Cited in.** Methods 2.5.1; Results 3.4.1.

A row is one analysis and set of metrics (6 rows).

The geometric-mean specificity ratio of the panel with a 90% interval (nested bootstrap that recomputes every metric inside each draw), for the earlier whole panel (100 to 103 metrics, including composition metrics that have a ratio of 1 by arithmetic) and for the 27 main-set metrics with a defined ratio (isosteric step: 1.075 [0.958, 1.178]).

Key columns.

Column	Meaning
analysis_set / metric_set / n_metrics	the analysis, the set of metrics and its size
geomean_sr / ci90_lo / ci90_hi / median_sr	geometric-mean ratio, 90% interval and median

Column	Meaning
n_clusters / status / note	resampling clusters, status and remarks

Excerpt (first 3 rows).

analysis_set	metric_set	n_metrics	geomean_sr	ci90_lo	ci90_hi
pilot set	earlier whole panel (all metrics ...	103	0.9642	0.7479	1.2195
main set	earlier whole panel (all metrics ...	102	0.9575	0.8969	1.0839
main set, packing-matched controls	earlier whole panel (all metrics ...	102	0.9947	0.9374	1.1342

Table S43. Re-predicted ladder: specificity ratio per step for all metrics

File. supplementary/S43_ladder_re_predicted_per_level.csv · 231 rows × 16 columns · public

Supports. Methods 2.5.1; Results 3.4.1. **Cited in.** Methods 2.5.1; Results 3.4.1.

A row is one metric at one step for one source (231 rows).

The re-predicted catalytic-lesion ladder per step (ALL steps pooled, isosteric, non-isosteric, Ala, Gly) for all 44 metrics, natural and de novo: pairs and systems, specificity ratio (median and mean) with intervals, and the verdict of the earlier rule.

Key columns.

Column	Meaning
source / metric_name / canonical_key / input_class / ame_flavour	the metric
level / n_pairs / n_systems	step and size
sr_median / sr_lo / sr_hi	ratio of medians with interval
sr_mean / sr_mean_lo / sr_mean_hi	ratio of means with interval
verdict_earlier_rule / applicability_status	verdict of the earlier rule and applicability status

Excerpt (first 3 rows where source = natural, level = isosteric, canonical_key = chai_ipm).

metric_name	level	n_pairs	n_systems	sr_mean	sr_mean_lo	sr_mean_hi
chai_ipm_max	isosteric	118	59	1.622	1.058	2.526
chai_ipm_mean	isosteric	118	59	1.963	1.44	2.707
chai_ipm_std	isosteric	118	59	1.046	0.605	1.878

Table S44. Second-shell lesion: every metric scored (Table 11 plus supplementary comparators and per-dose flags)

File. supplementary/S44_second_shell_lesion_all_metrics_ranked.csv · 37 rows × 59 columns · public

Supports. Results 3.4.2; the complete version of Table 11. **Cited in.** Methods 2.5.2; Results 3.4.2.

A row is one metric scored on second-shell removal (37 in all).

Table 11 with the 8 supplementary comparators and the flags at each dose (1, 2 and 4 residues removed): R with interval, the number of informative proteins, and whether the metric responds, whether the control response is above its noise and whether it is specific. No step counts for any metric, so no rank on this test is interpretable and no reference level is computed.

Key columns.

Column	Meaning
detection_group / rank_in_group / rank / rank_in_table_11	group by first dose detected, rank in the group, overall rank and rank among the 29 items of Table 11

Column	Meaning
metric_name / canonical_key / source / input_class / role / in_ranked_set	the metric and its role
n_steps / n_steps_interpretable	doses with a value and counted (0 for every metric)
r_second_shell_ / _lo / hi / n_second_shell	R, interval and informative proteins at dose 1, 2 and 4
responds_ / control_above_noise_ / specific_second_shell_	per-dose flags
r_mean_all_steps / first_step_detected / first_step_specific	mean R over all doses and the onsets

The shared block of 12 provenance columns (see Conventions) is also present and is not repeated above.

Excerpt (first 3 rows).

detection_group	rank_in_group	metric_name	r_mean_all_steps	n_steps_interpretable
second_shell_4	1	propka_catalytic_shift_mean	0.7005	0
second_shell_4	2	propka_catalytic_pka_mean	0.6915	0
never	1	catalytic_frac_buried	0.781	0

Notes.

- Because both arms are scored over the catalytic residues, a high raw R on this test measures location.

Table S45. Drift of the current Chai-1 engine against the earlier predictions

File. supplementary/S45_engine_drift_check.csv · 176 rows × 5 columns · public

Supports. Results 3.2. **Cited in.** Results 3.2.

A row is one metric on one re-run prediction (176 rows).

A drift check: four predictions (two cognate predictions of the substrate swap and two predictions of the catalytic-lesion ladder) were re-run with the current Chai-1 engine, and every metric is compared with its stored value. Most differences are small (median absolute difference 0.0007) but the largest is 0.91, which is why Chai-1 results are reported with seed noise and are not claimed to be bitwise reproducible.

Key columns.

Column	Meaning
prediction	the prediction that was re-run (system, condition and seed)
metric_name	the metric
stored / rerun / abs_diff	stored value, value of the re-run and their absolute difference

Excerpt (first 3 rows).

prediction	metric_name	stored	rerun	abs_diff
M0112_3dfr_a1_cognate_s101	ame_crystal_clash_max	2.722	2.7284	0.0064
M0112_3dfr_a1_cognate_s101	ame_crystal_pass	1.0	1.0	0.0
M0112_3dfr_a1_cognate_s101	ame_crystal_rmsd_max	0.8037	0.8055	0.0018

Table S46. Effective dimensionality of the panel

File. supplementary/S46_effective_dimensionality.csv · 4 rows × 10 columns · public

Supports. Results 3.4.1. **Cited in.** Results 3.4.1.

A row is one set of metrics and one panel (4 rows).

The participation ratio and related measures of the effective number of independent metrics, for the earlier whole panel and for the 29 structure-space main-set metrics, for native values (7.85 of 29) and for isosteric responses (10.5).

Key columns.

Column	Meaning
metric_set / panel / n_metrics / n_systems_complete	the set, the panel and sizes
participation_ratio / effective_rank_fraction	effective number of independent metrics and its share
var_explained_pc1 / n_pcs_for_90pct / median_abs_offdiag_r	variance in the first component, components for 90% and median absolute correlation between metrics

Excerpt (first 3 rows).

metric_set	panel	n_metrics	participation_ratio	var_explained_pc1
earlier whole panel	native values across systems	97	12.12	0.195
earlier whole panel	isosteric responses (catalytic ar...	76	14.2	0.155
main-set metrics (one member per ...	native values across systems	29	7.85	0.207

Table S47. De novo designs: burial equivalence of the created-site control

File. supplementary/S47_denovo_created_control.csv · 5 rows × 11 columns · public

Supports. Results 3.4.1. **Cited in.** Results 3.4.1.

A row is one burial metric (5 rows).

In the 30 de novo designs the control is a position mutated to the catalytic residue type. For the five burial metrics the table gives the catalytic-arm and control-arm medians, the paired difference with 90% interval and whether they differ.

Key columns.

Column	Meaning
metric / n_systems / n_clusters	the burial metric, designs and backbone clusters
catalytic_arm_median / control_arm_median	medians in the two arms
paired_diff_median / paired_diff_ci90_lo / paired_diff_ci90_hi / ratio_median	paired difference with interval and ratio
differs / note	1 if the arms differ

Excerpt (first 3 rows).

metric	n_systems	catalytic_arm_med an	control_arm_median	paired_diff_median	differs
catalytic_rel_sasa _mean	30	0.1433	0.1315	0.0278	0
catalytic_rel_sasa _max	30	0.2157	0.2042	0.0256	0
catalytic_sasa_mea n	30	32.1092	29.456	6.2169	0

Table S48. Experiments that could not be made or were retired

File. supplementary/S48_not_measurable.csv · 9 rows × 4 columns · public

Supports. Results 3.4.1. **Cited in.** Results 3.4.1.

A row is one experiment (9 rows).

The deformation experiment (apparent specific cells track the clash burden present before repacking), oxyanion-hole removal (the step that was run located no bound ligand in any of the 143 structures and removed a proxy residue, so it did not test the named change), metal removal (no cell could be scored on any metric), the de novo second shell and deformation experiments, PLACER on the isosteric step, the unrelated-protein sanity floor and the two trapping-mutant tiers.

Key columns.

Column	Meaning
reader_name / claim_type	the experiment and the kind of claim it would have supported
role	retired or not_measurable
mechanism_or_caveat	why it could not be made

Excerpt (first 3 rows).

reader_name	role	mechanism_or_caveat
sanity floor: unrelated protein	retired	signed paired-median statistic ca...
oxyanion-hole step	retired	well-defined perturbation, but no...
metal removal	not_measurable	no scoreable cell on any metric

Table S49. PLACER ensemble metrics on the isosteric step (not interpretable)

File. supplementary/S49_placer_isosteric.csv · 50 rows × 12 columns · public

Supports. Results 3.4.1. **Cited in.** Results 3.4.1.

A row is one PLACER metric in one analysis (50 rows).

The specificity ratio, verdict and applicability of 25 PLACER ensemble metrics on the isosteric step, in two analyses. They are not interpretable: the control arm is covered 1.6 to 3.2 times less than the catalytic arm in every crop configuration, so the count of specific metrics moves with the crop anchor alone.

Key columns.

Column	Meaning
analysis_set / metric_name / level / n_systems	the analysis, the metric, the step and enzymes
sr_median / sr_ci_lo / sr_ci_hi	specificity ratio with 95% interval
responds / specific / verdict / applicability_status	flags, verdict (blind, non_specific, specific, no_dynamic_range) and status (GLOBAL or X-CONFOUND)

Excerpt (first 3 rows).

analysis_set	metric_name	n_systems	sr_median	verdict	applicability_status
main set, packing-matched controls	placer_catalytic_rmsf_max	132	1.0106	blind	X-CONFOUND
main set, packing-matched controls	placer_catalytic_rmsf_mean	132	0.9058	blind	X-CONFOUND
main set, packing-matched controls	placer_centroid_dev_mean	132	0.9237	blind	X-CONFOUND

Appendix A. Main-text tables

The seven data tables of the Results (Tables 5 to 11), with the supplementary tables that hold the complete version or more detail. Tables 1 to 4 of the Methods are descriptive and are not released as files.

Table	File	Rows × columns	More detail in
T5	main/T5_dead_vs_active.csv	30 × 26	S11 , S7 , S8
T6	main/T6_plated_designs_auroc.csv	30 × 28	S13
T7	main/T7_substrate_swap.csv	30 × 32	S14 , S15
T8	main/T8_bglb_variants.csv	30 × 27	S16 , S17 , S18 , S19
T9	main/T9_plated_designs.csv	30 × 26	S21 , S20 , S23 , S22 , S24
T10	main/T10_specificity_catalytic_lesion.csv	35 × 36	S28 , S29 , S30
T11	main/T11_specificity_second_shell_lesion.csv	29 × 30	S44 , S29

Appendix B. Notes on the data

- Two sets of numbers for natural enzymes in the substrate swap: [S14](#) uses all 59 natural enzymes, Table 7 uses the evaluation set of 55.
- Two forms of the rank statistic R: [S36](#) is R on the 143 main enzymes with 90% intervals; Table 10 and [S28](#) are the pooled analysis on 195 enzymes with 95% intervals and the counted-step rule. Values for the same metric differ.
- Among the 29 prediction-based metrics of the ladder with ligand-distance-matched controls, the cells of [S3](#) keep the label NOT-RUN although the analysis was carried out (results in [S40](#) and [S41](#)).